

CS-4110: High Performance Computing with GPUs

Assignment 01

Objective:

This assignment aims to deepen your understanding of **CUDA programming** and **GPU acceleration** in High-Performance Computing (HPC). You will implement and analyze matrix operations on both **CPU and GPU**, compare execution times, and explore CUDA kernel configurations.

Questions:

1. CPU vs. GPU Performance Comparison

- (a) Implement **matrix multiplication** on both **CPU** and **GPU** using CUDA. A starter code is provided to you.
- (b) Measure execution times as well as data transfer times for **various matrix sizes** (e.g., **256×256**, **512×512**, **1024×1024**, **2048×2048**). Use cuda events to measure all the times.
- (c) Compute the **speedup factor** achieved by the GPU over the CPU.
- (d) **Present your results in a table and plot a suitable graph of:**
 - kernel execution time vs. matrix size
 - total execution time vs. matrix size
 - speedup vs matrix size
- (e) Provide your reflection of all the presented results.

2. Result Verification

- (a) Verify the **correctness** of GPU matrix multiplication by comparing it against the CPU results.
- (b) Implement a function to check the absolute difference between CPU and GPU outputs (error tolerance of **1e-4**).
- (c) If discrepancies exist, explain possible reasons (e.g., precision errors, memory access issues).

3. 1D vs. 2D Kernel Launch Configurations

- (a) Implement **two different CUDA kernel launches** for matrix multiplication:
 - **1D grid (linearized indexing)**
 - **2D grid (block and thread indexing in x & y dimensions)**
- (b) Compare their execution times for different matrix sizes.

(c) **Which configuration is more efficient?** Justify your answer.

4. Matrix multiplication on two different GPUs.

Compare the performance of the matrix multiplication on the GPU provided to you for this course as well as the one provided by Google in the Colab. Provide a reflection on your comparison.

Submission Guidelines:

- Submit **well-commented source code** for both **CPU and GPU implementations**.
- The code should be able to compile and execute successfully out of the box.
- Include a **report (3-5 pages)** discussing:
 - Implementation details
 - Performance analysis (tables & graphs)
 - Comparison of **CPU vs. GPU execution time & speedup**
 - Observations on **1D vs. 2D kernel configurations**
 - Observations on **different GPU results**
- Submit all your work in a **single zip file** with the name same as your registration number (e.g. **23i-1234-a1.zip**). Inside this zip you have one directory for **report** and the other directory **src** which contains the complete source code with makefile. There should be a **results** directory containing the spreadsheet which you used to make tables and plot graphs.